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Academic Year (JAN-JUNE 2026)

Integrated M.Sc. IT. (SEM-IV)

<table>
<tr><td>Course Code</td><td colspan="4">150020417</td></tr>
<tr><td>Category</td><td colspan="4">SEC</td></tr>
<tr><td>Course Title</td><td colspan="4">Business Analysis Using R</td></tr>
<tr><td rowspan="2">Scheme and Credits</td><td>Total Lectures</td><td>Lec</td><td>Tutorial</td><td>Credits</td></tr>
<tr><td>36</td><td>36</td><td>0</td><td>2</td></tr>
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</table>

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Detailed Schedule	
Day 1-3 CO-1	
1	**Introduction**: Broad areas (classification) of Statistics, Applications of Statistics in Computer science
2	**Data**: Data and information, Types of Data : Qualitative (Nominal, Ordinal) and Quantitative (Discrete, continuous)
3	**Descriptive Statistics:** Central Tendency, Mean and its Characteristics, Median and its Characteristics, Mode and its Characteristics
4	Range, Quartiles (Q1, Q2, Q3), Inter quartile Range (IQR), Percentile.
5	Introduction to R Studio: What is R, Why use R,
Essential Assignment	
1	Create a vector X with dataset (99,86,87,88,111,103,87,94,78,77,85,86) and calculate the following: 1)Mean 2) Mode 3) Median 4) Range 5) Quartiles 6) IQR
2	Use R to answer the following questions. Calculate sqrt(100 – 3* 4.52) If x = 3, y= 2 and z = -5 what is the value of x3 – 2xy2 + 3z + z2?
3	Create file (data.csv) using following data 12,15,17,18,14,12,6,23,22,12,17,18,15,20,25 Calculate mean , Median, Mode, Range, Q1,Q3 and IQR manually as well as using separate command in R.
4	Create a csv file for the dataset given in question 1 and write a program to calculate mean, Median, quartiles and IQR.
Desirable Assignment	
1	Use R to answer the following questions. Write down the answers. a. Generate a numbers from 1-100. b. If x = 5, y= 3 and z = -4 what is the value of $x^2 – 2xy + 3z + z^2$?
2	Create file (data.txt) using following data 33.5 ,57.1 ,49.7 ,40.2, 44.2, 45.2, 47.8, 38.0 ,53.9 ,41.1 ,41.7, 40.8 ,55.5 ,43.5 ,49.1 Write a program in R to calculate mean, Median, quartiles and IQR for given data.

Course Code: 150020417

Detailed Schedule	
Day 4-6 CO-2	
1	**Describing Data Visually:** Frequency Distributions and Histograms, Pie Charts, Bar Charts, Pareto Chart, Scatter Plots (Degree of Association); Line Charts, Box Plots
2	**Dispersion**: Variance, Standard Deviation and its Characteristics, Coefficient of Variation
3	**Skewness**: Definition and its types, Measures of Coefficient of Skewness
4	A bowler's scores for six games were 182, 168, 184, 190, 170, and 174. Using these data as a sample, compute the following descriptive statistics: a. Range c. Standard deviation b. Variance d. Coefficient of variation
Essential Assignment	
1	Create a vector X with data set (1, 2, 3, 4, 5) and vector Y with data set (3, 7, 8, 9, 12). Create line chart and scatter chart in R.
2	Write a program in R to create pie chart for vector (10, 20, 30, 40) with appropriate label and colors.
3	Create file (data.csv) using following data 33.5 ,57.1 ,49.7 ,40.2, 44.2, 45.2, 47.8, 38.0 ,53.9 ,41.1 ,41.7, 40.8 ,55.5 ,43.5 ,49.1 Write a program in R to calculate variance, standard deviation and coefficient of variation for given data.
4	Create a vector X with data set ("A","B","C","D") and vector Y with data set (2, 4, 6, 8). Create horizontal and vertical Bar chart in R.
5	Write a program in R to create histogram for vector **(19, 23, 11, 5, 16, 21, 32,14, 19, 27, 39)**
6	Write a program to create box plot for mtcars dataset available in R.
Desirable Assignment	
1.	Create a vector X with data set (5,7,8,7,2,2,9,4,11,12,9,6) and vector Y with data set (99,86,87,88,111,103,87,94,78,77,85,86). Create line chart and scatter chart in R with appropriate labels and color coding.
2.	Use file shopping data.csv available in kaggle.com to create Line, scatter and pie chart in R. (https://www.kaggle.com/carrie1/ecommerce-data)

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Detailed Schedule

Day 7-8 CO-3

1	**Measures of Association:** Covariance, Correlation, Coefficient of Correlation
2	**Regression:** Introduction, Independent and dependent data, Simple Regression Analysis, Least Square method, Fitting of regression line (Y On X and X on Y) Prediction of dependent data using regression line
3	For the following Data

X	2	3	5	1	8
Y	25	25	20	30	16

Calculate

1. Calculate correlation and interpret
2. Calculate covariance
3. Develop a Scatter diagram for this data.
4. Compute the estimated regression equation.
5. Predict value of Y for X=4
6. Predict value of X for Y=23

Essential Assignment

1. Create vector Values of height
151, 174, 138, 186, 128, 136, 179, 163, 152, 131
And Values of weight.
63, 81, 56, 91, 47, 57, 76, 72, 62, 48
Write a program to compute regression equation Y on X and predict value of Y for X=170.

2. Write a program in R to calculate the following for given dataset.

X	2	3	5	1	8
Y	25	25	20	30	16

1. Calculate correlation and interpret
2. Calculate covariance
3. Develop a Scatter diagram for this data.
4. Compute the estimated regression equation.
5. Predict value of Y for X=4
6. Predict value of X for Y=23

3. Write a program in R to create csv file for given data set and calculate the following:

Salary	110	130	140	160	170	90	100	130	150
Bonus	12	15	14	15	20	12	10	12	18

1. Calculate correlation and interpret
2. Calculate covariance
3. Develop a Scatter diagram for this data.
4. Compute the estimated regression equation.
5. Predict value of Bonus for salary=105
6. Predict value of Salary for Bouns=14

Desirable Assignment	
1.	For the kaggle dataset California housing prices (https://www.kaggle.com/camnugent/california-housing-prices) apply linear regression. Decide independent and dependent variable and plot the scatter chart.
2.	For the kaggle dataset Breast cancer (https://raw.githubusercontent.com/GTPB/PSLS20/master/data/breastcancer.csv) apply linear regression. Decide independent and dependent variable and plot the scatter chart.

Detailed Schedule	
Day 9-11 CO-2	
1	**Unit 3: Distribution, Distribution functions and Measure of Central Tendency** **Introduction:** Common Framework: Experiment, Event, Elementary Events, Sample Space;
Essential Assignment	
1.	**Generating random numbers.** Set your seed to 1 and generate 10 random numbers using runif and save it in an object called random_numbers
2.	Using the function ifelse and the object random_numbers simulate coin tosses. Hint: If random_numbers is bigger than .5 then the result is head, otherwise is tail.
4.	
Desirable Assignment	
1.	A physical therapist at Gujarat Technological University knows that the football team will play 40% of its games on artificial turf this season. He also knows that a footbal player's chances of incurring a knee injury are50% higher if he is playing on artificia turf instead of grass. If a player's probability of knee injury on artificial turf is 0.42, wha is the probability that: I. A randomly selected football player incurs a knee injury? II. A randomly selected football player with a knee injury incurred the injury playing on grass?

Detailed Schedule	
Day 12-14 CO-4	
1	Conditional Probability; Mutually Exclusive Events, Independent Events
2	Discrete Probability Distributions: Introduction, Binomial Distribution, Poisson Distribution, Applications
3	Assuming that half of the Indian population is vegetarian. Estimate how many investigators out of 100 will report that 3 or less are vegetarian in the sample of 10 individuals.
4	In a city of some western country 70 % of the married persons take divorce. What is The Probability that at least three among four persons will take divorce?
5	Phone calls arrive at the rate of 48 per hour at the reservation desk for Regional airways. (1) Compute the probability of receiving three calls in a five minute interval of time. (2) Compute the probability of receiving exactly ten calls in fifteen minutes. (3) If no calls are being processed, what is the probability that the agent can take three minutes for personal time without being interrupted by a call?
6	For two mutually events A and B having P(A)=0.3 and P(B)=0.4, compute P(AUB) and P(A|B).
Essential Assignment	
1.	Suppose there are twelve multiple choice questions in an English class quiz. Each question has five possible answers, and only one of them is correct. Find the probability of having 1) exactly four and 2) four or less correct answers if a student attempts to answer every question at random.
2.	If there are twelve cars crossing a bridge per minute on average, find the probability of having seventeen or more cars crossing the bridge in a particular minute
3.	Write a program in R to solve Q4 and Q5.
4.	Usage of **dbinom(),pbinom(),qbinom()and rbinom() method of R**
Desirable Assignment	
1.	Survey shows that 40% of the students are using 5G mobile. In a random sample of 10 students, what is the probability that two students have 5G mobile? (write a program in R)
2.	A typist in a company commits the following number of mistakes per page in typing 432 pages. Does this information verify that the mistakes are distributed according to Poisson law?

No. of mistakes/Page	0	1	2	3	4	5
No. of pages	223	142	48	15	4	0

Write a program in R.

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Detailed Schedule	
Day 15-17 CO-1	
1	**Continuous Probability Distributions:** Introduction, Normal Distribution, Exponential Distribution, Application
2	A Population has a mean of 200 and a standard deviation of 50. Suppose a simple random sample of size 100 is selected and x is used to estimate μ. (1) What is the probability that the sample mean will be within ± 5 of the population mean? (2) What is the probability that the sample mean will be within ± 10 of the population mean?
3	Assume that admission test scores are normally distributed with mean 400 and $\sigma = 100$ marks. Find the probability of scores between 300 and 450.
4	If the probability that a fluorescent light has a useful life of at least 500 hours is 0.85, find the probability that among 20 such lights • At least 18 will have a useful life of at least 500 hours • Exactly 15 will have a useful life of at least 500 hours • None will have a useful life of at least 500 hours
Essential Assignment	
1.	Usage of **dnorm(),pnorm(),qnorm()and rnorm() method of R**
2.	Assume that the test scores of a college entrance exam fits a normal distribution. Furthermore, the mean test score is 72, and the standard deviation is 15.2. What is the percentage of students scoring 84 or more in the exam?
3.	Suppose the mean checkout time of a supermarket cashier is three minutes. Find the probability of a customer checkout being completed by the cashier in less than two minutes.
4.	Write a program in R to solve Q2.
5.	Write a program in R to solve Q4.
Desirable Assignment	
1.	Marks of large number of students are distribution normally with mean 50 and standard Deviation 12. If a student is selected at random what is the probability that his marks will be Between (1) 38 and 62 (2) 26 and 74 (Given area between z = 0 and z = 1 is 0.34135 & z = 0 and z = 2 is 0.47725
2.	A machinery company has received a big order to produce electric motors for a manufacturing company. In order to fit in its bearing, the drive shafted the motor must have a diameter of 5.1 + 0.05 (inches). The company's purchasing agent realizes tha there is a large stock of steel rods in inventory with a mean diameter of 5.07'', and a standard deviation of 0.07''. What is the probability of a steel rod from inventory fitting the bearing?

Detailed Schedule	
Day 18-20 CO-2	
1	**Types of Sampling:** Random, Non random;
2	Sampling Distribution of x-bar; Central Limit Theorem;
3	z Formula for Sample Mean; Standard Error of Mean; Sampling from a Finite Population
4	When a sample of 70 retail executives was surveyed regarding the poor November performance of the retail industry, 66 percent believed that decreased sales were due to unreasonably warm temperatures, resulting in consumers' delaying purchase of cold-weather items. Construct a 95 percent confidence interval for the true proportion.
5	A sample of items selected from normal population is 10, 5, 7, 8, 20, 25, 15, 2 and 12. Compute point estimate and 95% interval estimate of population mean.
Essential Assignment	
1.	Write a program in R to generate sample data from the vector (23,45,21,34,5,6,7,8,86,45,3)
2.	Write a program in R to generate sample data from the vector using replacing (23,45,21,34,5,6,7,8,86,45,3)
3.	Write a program in R to generate sample data from the vector using uneven probabilities (23,45,21,34,5)
4.	Write a program in R to perform random sampling of data frames rows using sample function.
5.	For the kaggle dataset Breast Cancer (https://www.kaggle.com/uciml/breast-cancer-wisconsin-data) uses any two columns and apply sample function to generate any 10 random samples.
Desirable Assignment	
1.	Write a program in R to perform random sampling of list of elements using sample function.
2.	For the kaggle dataset California housing prices (https://www.kaggle.com/camnugent/california-housing-prices) use any two columns and apply sample function to generate any 10 random samples with replacement.

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Detailed Schedule	
	Day 21-23 CO-3
1	Estimating the Population Mean using z Statistic (σ Known): finite correction factor
2	Estimating the Population Mean using the z Statistic when the Sample Size is Small
3	Construct Z confidence interval for mean
4	A survey of 611 office workers investigated telephone answering practices including how often each office worker was able to answer incoming telephone calls and how often incoming telephone calls went directly to voice mail. A total 281 office workers indicated that they never need voicemail and are able to take every telephone call. 1. At 90% confidence, what is margin of error? 2. What is the 90% confidence interval for the proportion of the population of office workers who are able to take every telephone call?
5	For a random sample of 36 items and a sample mean of 211, compute a 95% confidence interval of mean if the population standard deviation is 23.
	Essential Assignment
1.	Write a program in R to generate population of 100 random numbers. Take 80 samples out of population and calculate 95% confidence interval.
2.	Write a program in R to calculate 88% confidence interval μ=5, sd=2 and n=20.
3.	Write a program in R to generate 36 random samples and calculate 90%, 95% and 99% confidence interval.
4.	For the kaggle dataset California housing prices (https://www.kaggle.com/camnugent/california-housing-prices) use any two columns and apply sample function to generate any 10 random samples with replacement and calculate 75% confidence interval.
	Desirable Assignment
1.	For the kaggle dataset accident (https://www.kaggle.com/rajanand/accidents-in-india/version/1) column Total and apply sample function to generate any 80 random samples without replacement and calculate 90% confidence interval.(write a program in R)
2.	For the kaggle dataset Heart disease (https://www.kaggle.com/cdabakoglu/heart-disease-classifications-machine-learning/data) for any two columns calculate 90% and 95% confidence interval. (Write a program in R)

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Detailed Schedule	
Day 24-26 CO-3	
1	Estimating the Population Mean using t Statistic (σ Unknown): The t distribution, Characteristics of t distribution, Reading the t distribution table
2	Confidence intervals to estimate the population mean using t statistic.
3	For random samples of 18 managers and measures the amount of extra time they work during a specific week and obtains the results shown below 6,21,17,20,7,0,8,16,29,3,8,12,11,9,21,25,15,16 Construct 90% confidence interval to estimate the average amount of extra time per week worked by manager.
4	Suppose the following data are normally selected from a population of normally distributed values 50,40,51,43,48,44,57,54,47,39,42,48,45,39,43,46 Construct a 95% confidence interval to estimate the population mean.
Essential Assignment	
1.	Write a program in R to create a vector (1,21,17,20,7,0,8,16,29,3,8,12,11,9,21,25,15,16) and calculate 88% interval estimate using t distribution.
2.	Write a program in R to read any 10 integer number from user using scan() and calculate 95% interval estimate using t distribution.
3.	Write a program in R to read data.csv file and calculate 90% and 99% interval estimate using t distribution.
4.	For the kaggle dataset California housing prices (https://www.kaggle.com/camnugent/california-housing-prices) use any two columns and calculate 80% confidence interval using t distribution.
Desirable Assignment	
1.	Write a program in R to use the kaggle dataset accident (https://www.kaggle.com/rajanand/accidents-in-india/version/1). Use column named Total and apply sample function to generate any 80 random samples without replacement and calculate 97.5% confidence interval using t distribution.
2.	Write a program in R to use the kaggle dataset Heart disease (https://www.kaggle.com/cdabakoglu/heart-disease-classifications-machine-learning/data). Use any two columns and calculate 88.5% and 76% confidence interval using t distribution.

Detailed Schedule	
Day 27-29 CO-1	
1	Hypothesis Testing : Null Hypothesis, Alternate Hypothesis; Type I & Type II Errors
2	Testing Hypotheses about a Population Mean using z Statistic (σ Known); Using Critical Value Method to test Hypotheses
3	Consider the following hypothesis test: H 0 : sample mean=15, H a : sample mean$\neq$15 A sample of 50 provided a sample mean of 14.15. The population standard deviation is 3. (1) Compute the value of the test statistic (2) What is the p- value? (3) At α=0.05, what is your conclusion? (4) What is the rejection rule using the critical value? What is your conclusion?
4	Company manufactures car tyres. Mean life of tyres is 42000 km with a standard deviation of 3000 km. Company changes the production process to improve the quality. After this change, a test sample of 20 new tyres has a mean life of 43500 km with same s.d. as before. Do you think that the new car tyres are significantly superior to the earlier one?
Essential Assignment	
1.	Write a program in R to test null hypothesis using Z test where populations mean=3, σ = 2 and sample given is (3, 7, 11, 0, 7, 0, 4, 5, 6, 2). If null hypothesis is H0 : sample mean=3, what is your conclusion.
2.	Suppose the IQ in a certain population is normally distributed with a mean of μ = 100 and standard deviation of σ = 15. A scientist wants to know if a new medication affects IQ levels, so she recruits 20 patients to use it for one month and records their IQ levels at the end of the month. Data are **(88, 92, 94, 94, 96, 97, 97, 97, 99, 99, 105, 109, 109, 109, 110, 112, 112, 113, 114, 115)** Write a program in R to shows how to perform a one sample z-test using 99% confidence interval to determine if the new medication causes a significant difference in IQ levels.
3.	Consider the following hypothesis test: H 0 : sample mean=15, H a : sample mean$\neq$15 A sample of 50 provided a sample mean of 14.15. The population standard deviation is 3.Write a program in R to shows how to perform a one sample z-test using 95% level of significance.
4.	Write a program in R to test the hypothesis that the mean systolic blood pressure in a certain population equals 140 mmHg. The standard deviation has a known value of 20 and a data set of 55 patients is available as (use scan to insert data) (120,115,94,118,111,102,102,131,104,107,115,139,115,113,114,105,115,134,109,109,93,118,109,106,125,150,142,119,127,141,149,144,142,149,161,143,140,148,149,141,146,159,152,135,134,161,130,125, 141,148,153,145,137,147,169)

Desirable Assignment	
1.	Explain Type I and Type II error with example
2.	ABC company manufactures tablets. For quality control, two sets of tablets were tested. In the first group, 32 out of 700 were found to contain some sort of defect. In the second group, 30 out of 400 were found to contain some sort of defect. Is the difference between the two groups significant? Use a 5% alpha level.(write a program in R)
3.	We have two groups of student A and B. Group A with an early morning class of 400 students with 342 female students. Group B with a late class of 400 students with 290 female students. Use a 5% alpha level. We want to know, whether the proportions of females are the same in the two groups of the student? (write a program in R)

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Detailed Schedule

Day 30-32 CO-4

1	Population Mean Testing Hypotheses about a Mean using t Statistic (σ Unknown)
2	The following data shows sales made by salespeople from two different cities. City A: 59,68,44,71,63,46,69,54,48 City B : 50,36,62,52,70,41 Assuming the populations sampled to be approximately normal having the same variance, test whether there is any significant difference between the means of these samples.
3	The data below are a random sample of 9 firms chosen from the "Digest of Earnings Reports" in The Wall Street Journal on September 2011. 1. Find the mean change in earnings per share between 2010 and 2011. 2. Find the standard deviation of the change and the standard error of the mean. 3. Were average earnings per share different in 2010 and 2011? Test at $\alpha = 0.02$.

Firm	1	2	3	4	5	6	7	8	9
2010 earnings	1.38	1.26	3.64	3.50	2.47	3.21	1.05	1.98	2.72
2011 Earnings	2.48	1.50	4.59	3.06	2.11	2.80	1.59	0.92	0.47

Essential Assignment

1.	Write a program in R to test null hypothesis using t test for given dataset at 95% level of significance. Conclude your results.

Women's Height	1.38	1.26	3.64	3.50	2.47	3.21	1.05	1.98	2.72
Men's Height	2.48	1.50	4.59	3.06	2.11	2.80	1.59	0.92	0.47

2.	Suppose you are a company producing cookies. Each cookie is supposed to contain 10 grams of sugar. The cookies are produced by a machine that adds the sugar in a bowl before mixing everything. You believe the machine does not add 10 grams of sugar for each cookie. If your assumption is true, the machine needs to be fixed. You stored the level of sugar of thirty cookies. Write a program in R to create a distribution with 30 observations with a mean of 9.99 and a standard deviation of 0.04.Apply t test to check the following **H0**: The average level of sugar is equal to 10 **H1**: The average level of sugar is different than 10
3.	Write a program in R to apply t test on mtcras data set and conclude your results.

Desirable Assignment

1.	Nine computer-components dealers in major metropolitan areas were asked for their prices in $ on two similar colour inkjet printers. The results of this survey are given below. At $\alpha = 0.05$, is it reasonable to assert that, on average, the Apson printer is less expensive than the Okaydata printer? (Write a program in R)

Dealer	1	2	3	4	5	6	7	8	9
Apson	250	319	285	260	305	295	289	309	275
Okaydata	270	325	269	275	289	285	295	325	300

<table>
<tr><th colspan="2">Detailed Schedule</th></tr>
<tr><td colspan="2">Day 33-34 CO-2</td></tr>
<tr><td>1</td><td>Testing Hypotheses about a Proportion</td></tr>
<tr><td>2</td><td>In a sample of 500 people from a village, 280 are found to be rice eaters and the rest are wheat eaters. Can we assume that both the food articles are equally popular?</td></tr>
<tr><td>3</td><td>In a hospital 480 female and 520 male babies were born in a week. Do these figures confirm the hypothesis that males and females are born in equal numbers?</td></tr>
<tr><td>4</td><td>Students Safe Driving has targeted seat-belt usage as a positive step to reduce accidents and injuries. Before a major campaign at one college, 44 percent of 150 drivers entering the college parking lot were using their seat belts. After the seat-belt awareness program, the proportion using seat-belts had risen to 52 percent in a sample of 200 vehicles. At a 0.04 significance level, can the students conclude that their campaign was effective?</td></tr>
<tr><td colspan="2">Essential Assignment</td></tr>
<tr><td>1.</td><td>Write a program in R to apply Z test for a population of mice containing half male and have female (p = 0.5 = 50%). Some of these mice (n = 160) have developed a spontaneous cancer, including 95 male and 65 female.</td></tr>
<tr><td>2.</td><td>We have two groups of individuals Group A with lung cancer x=490: n = 500 and Group B, healthy individuals x=400: n = 500. Write a program in R to test whether the proportions of smokers are the same in the two groups of individuals?</td></tr>
<tr><td>3.</td><td>Use the built-in R data set named PlantGrowth. It contains the weight of plants obtained under a control and two different treatment conditions. Write a program in R to apply one way ANOVA and interpret the results.</td></tr>
<tr><td>4.</td><td>Use data on patients having stomach, colon, ovary, brochus, or breast cancer. The objective of the study was to identify if the number of days a patient survived was influenced by the organ affected. Our dependent variable is Survival measured in days. Our independent variable is Organ. The data is available here http://lib.stat.cmu.edu/DASL/Datafiles/CancerSurvival.html and a cancer-survival . Write a program in R to apply one way ANOVA and interpret the results.</td></tr>
<tr><td colspan="2">Desirable Assignment</td></tr>
<tr><td>1.</td><td>The following table gives the yields on 15 sample fields under three varieties of seeds.
<table>
<tr><th colspan="3">Yields</th></tr>
<tr><td>A</td><td>B</td><td>C</td></tr>
<tr><td>5</td><td>3</td><td>10</td></tr>
<tr><td>6</td><td>5</td><td>15</td></tr>
<tr><td>8</td><td>2</td><td>7</td></tr>
<tr><td>1</td><td>10</td><td>13</td></tr>
<tr><td>5</td><td>0</td><td>17</td></tr>
</table>
Write a program to compute One-Way ANOVA table and find the value of F at 5% level of significance.</td></tr>
</table>

2.	National Transportation Safety Board (NTSB) wants to examine the safety of compact cars, midsize cars, and full-size cars. It collects a sample of three for each of the treatments (cars types). Using the hypothetical data provided below, test whether the mean pressure applied to the driver's head during a crash test is equal for each types of car. Compact Car | Mid size car | Full Size car 643 | 469 | 486 655 | 427 | 456 702 | 525 | 402 Write a program to compute One-Way ANOVA table and find the value of F at 5% level of significance.

Compact Car	Mid size car	Full Size car
643	469	486
655	427	456
702	525	402

Detailed Schedule

Day 35 CO-1

1	**Analysis of variance:** One way analysis of Variance
2	The following table gives the yields on 15 sample fields under three varieties of seeds. Compute One-Way ANOVA table and find the value of F at 5% level of significance.
3	National Transportation Safety Board (NTSB) wants to examine the safety of compact cars, midsize cars, and full-size cars. It collects a sample of three for each of the treatments (cars types). Using the hypothetical data provided below, test whether the mean pressure applied to the driver's head during a crash test is equal for each types of car.

Yields

A	B	C
5	3	10
6	5	13
8	2	7
1	10	13
5	0	17

Compact Car	Mid size car	Full Size car
643	469	486
655	427	456
702	525	402

Write a program to compute One-Way ANOVA table and find the value of F at 5% level of significance.

Essential Assignment

1. The Following table shows the sample values of three independent normal random variables, X1, X2 and X3. Assuming that they have equal variances, test the hypothesis that they have the same mean by using ANOVA. ($F_{9,2}$=19.385)

X1	13	11	16	22
X2	16	8	21	11
X3	15	12	25	10

2. Three brands of battery are under study. It is suspected that the life of 3 brands is different. 5 batteries of each brand are tested with the following results.

Brand 1	Brand 2	Brand 3
10	6	8
6	8	4
2	7	6
6	4	8
2	2	4

Assuming that they have equal variances, test the hypothesis that they have the same mean by using ANOVA. ($F_{12,2}$=19.40)

Detailed Schedule	
Day 36	
1	Revision and doubt solving session
Essential Assignment	
1.	Submit hard copy of theory assignments